

Introduction to Cosmology

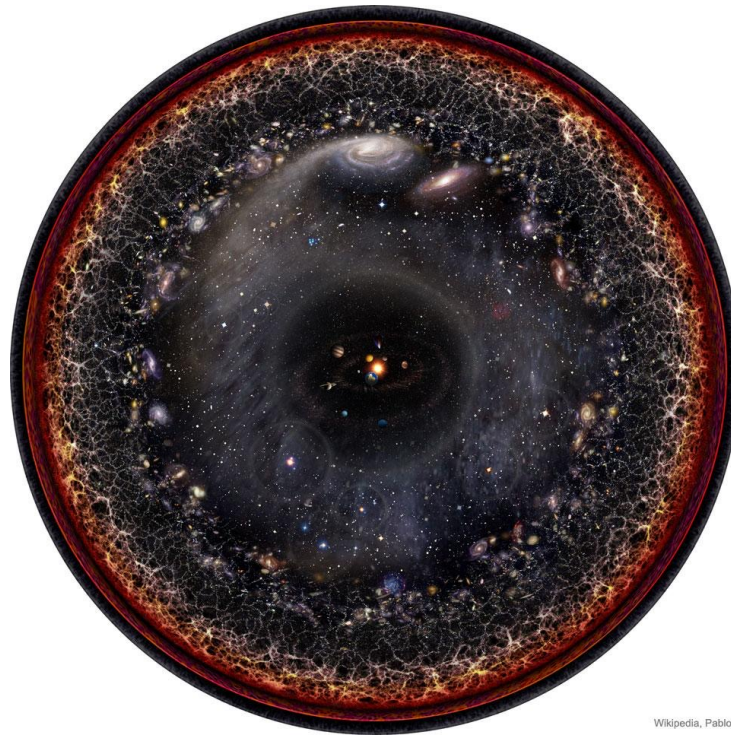
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Hunter College May 10th 2023

What is Cosmology?

The Study of the Universe (Cosmos) at the Largest Scales

The (Observable) Universe

Different from the
Actual Universe!



Age: 13.8 Billion Years

Earth's Age: 4.5 Billion Years ($\sim 1/3$ Universe's)

96 Billion Lightyears
 $\sim 10^{27}$ m

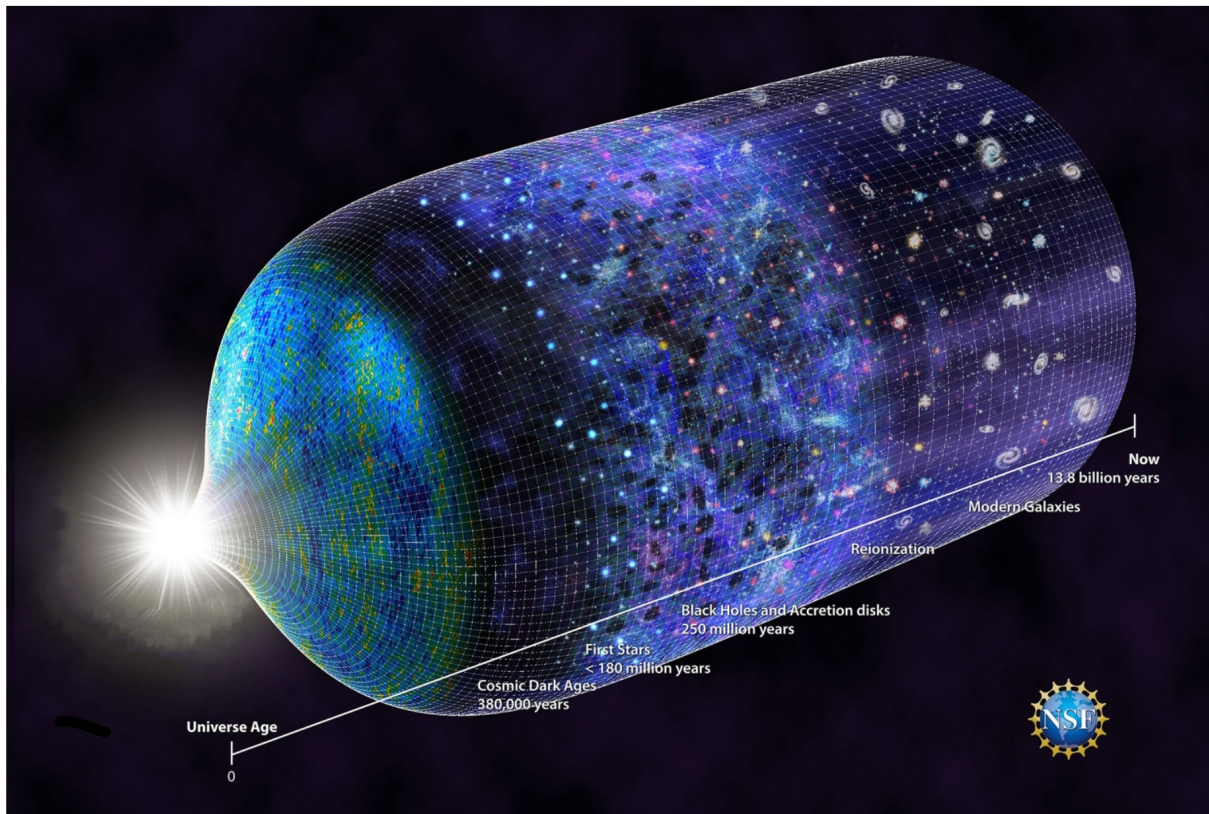
Diameter of Milky Way:
0.1 Million Lightyears
 $\sim 10^{21}$ m
(Million Times Smaller)

Dimensional Analysis: A Million Galaxies?

I reason that since the Universe is a million times larger than the Milky Way, then there can no more than a million galaxies in the Universe. Am I right?

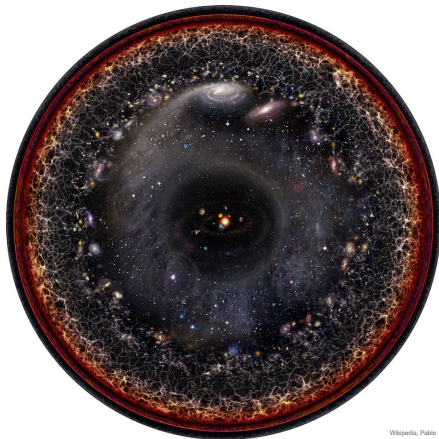
No! The Universe is a 3-dimensional space, not a (1-dimensional) line. There are an estimated ~ 100 billion galaxies in the Universe.

A Dynamical Universe



The Universe Expands!

How do we know these statements?

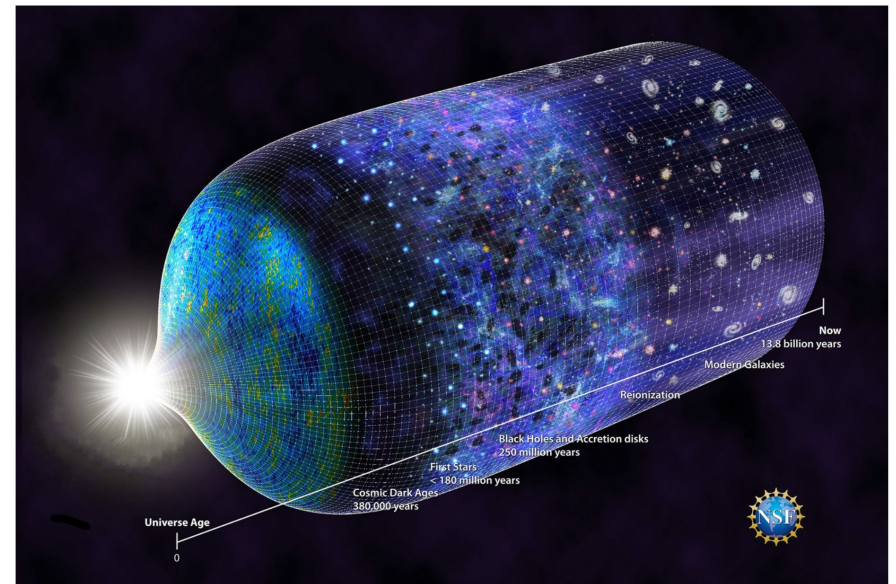


Wikipedia, Pablu



Size: 96 Billion Lightyears

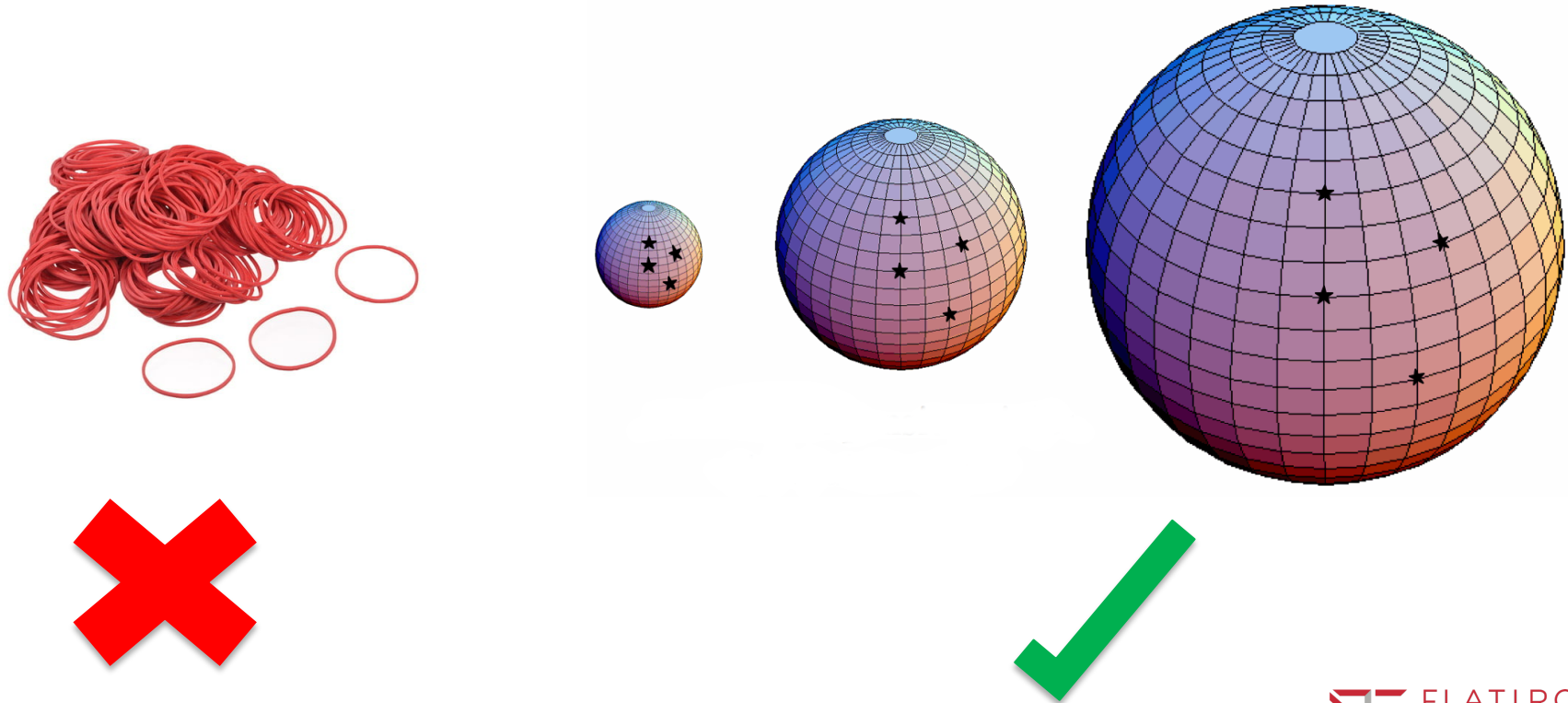
Age: 13.8 Billion Years



The Universe Expands!

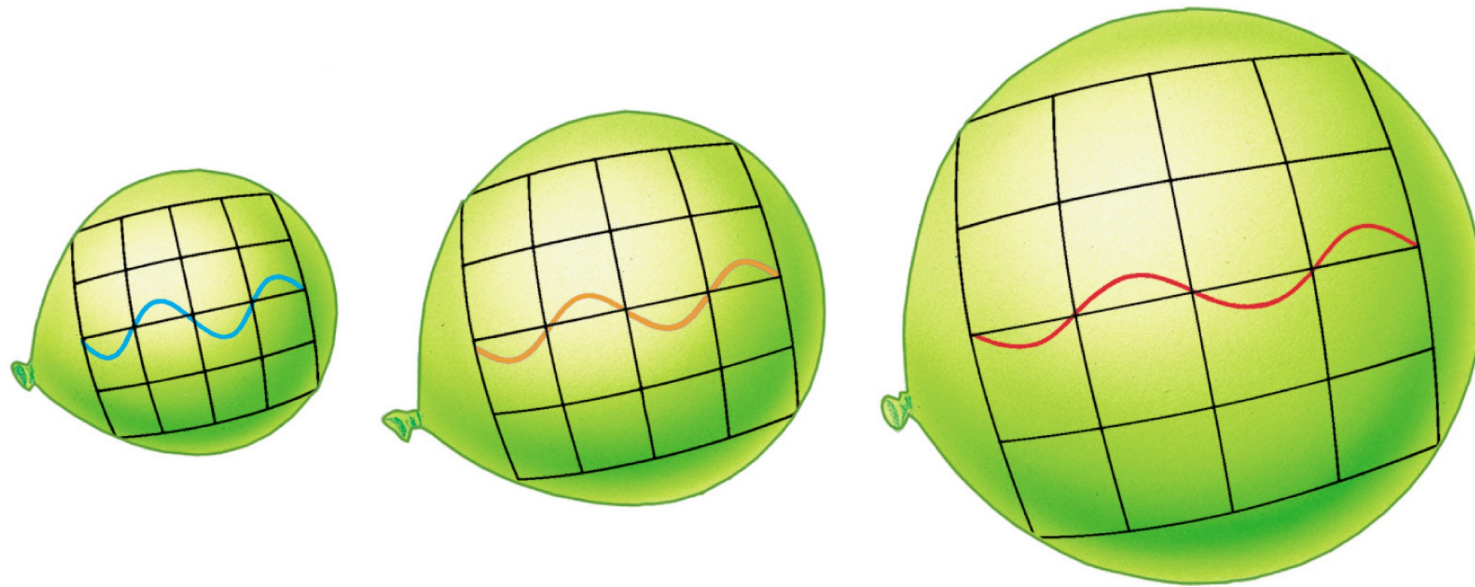
And what else do they predict?

What does it mean for a Universe to expand?



What does it mean for a Universe to expand?

While point particles remain unchanged, the wavelength of waves expand (redshift)!



What does it mean for a Universe to expand?

Fundamentally, what does it mean to say that our universe is expanding?

- A) Galaxies are moving apart through space
- B) Space itself is expanding
- C) Everything is expanding, including galaxies star systems, and star clusters

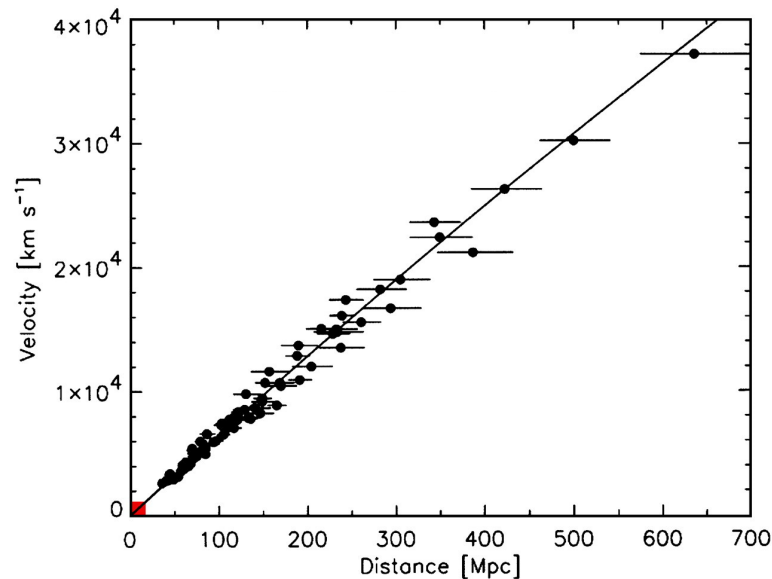
Can I achieve superluminal speeds?

Maybe I was wrong about the number of galaxies, but now I reason that since the Universe is expanding, then galaxies further away move faster away. If I then look far enough, then I can then see a galaxy move away faster than the speed of light. Am I right?

Yes! In an expanding Universe,
apparent superluminal speed propagation is okay.

How does one tell the Universe expands?

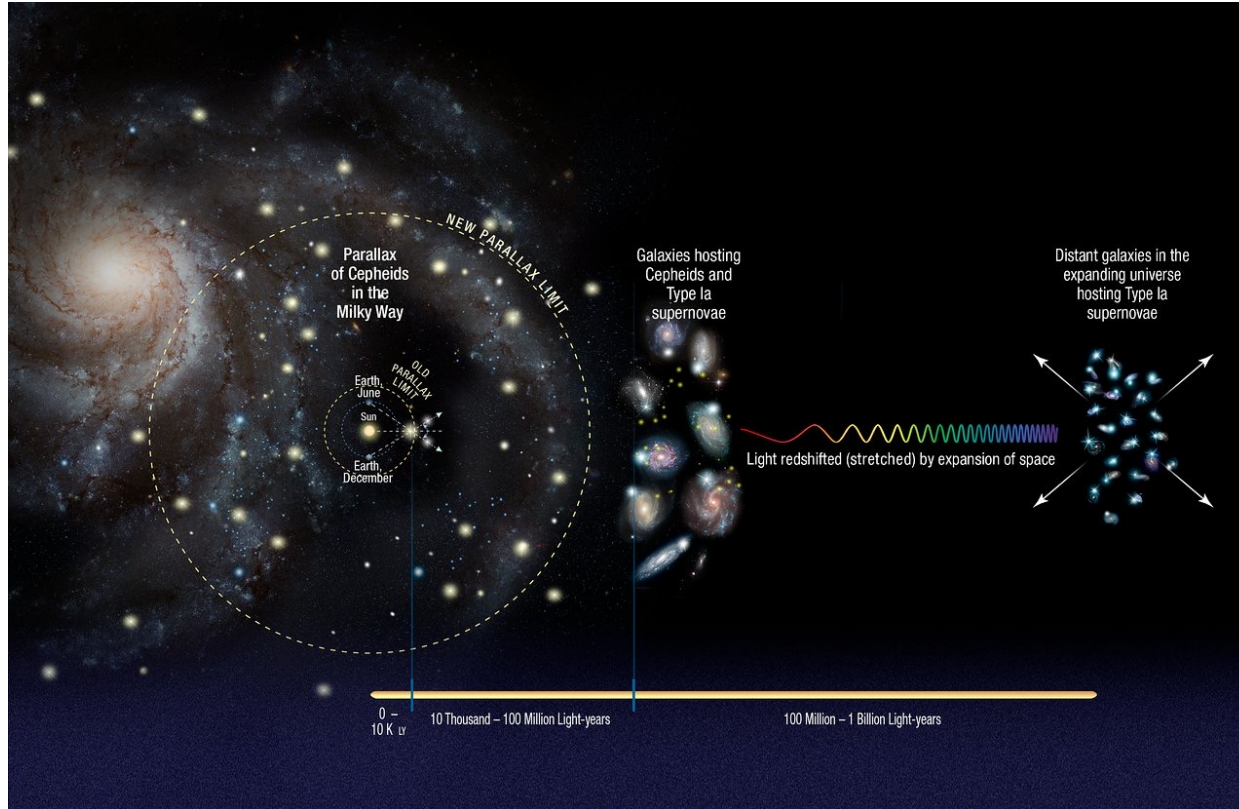
Historical Method: Galaxy Recession (Hubble-Lemaître Law, 1927)



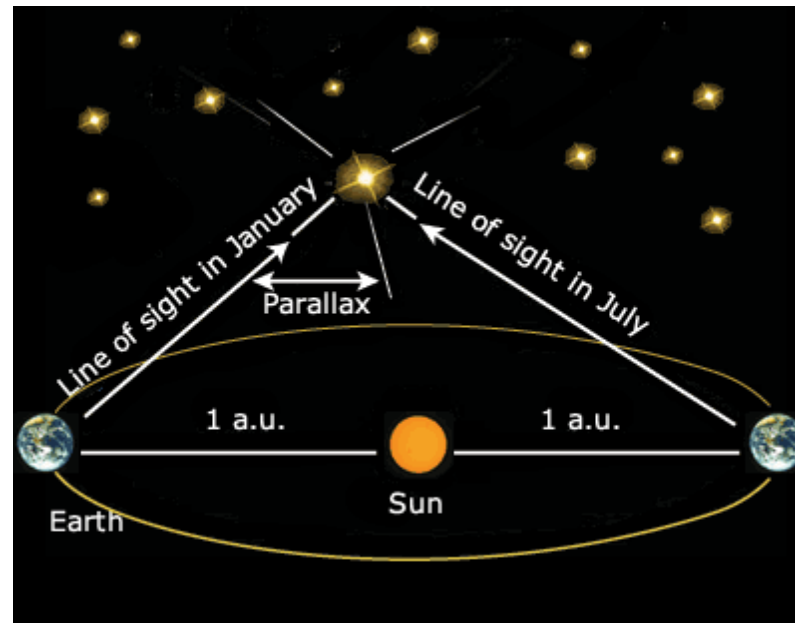
$$v = H_0 d$$

Modern Analog: Cosmic Distance Ladder

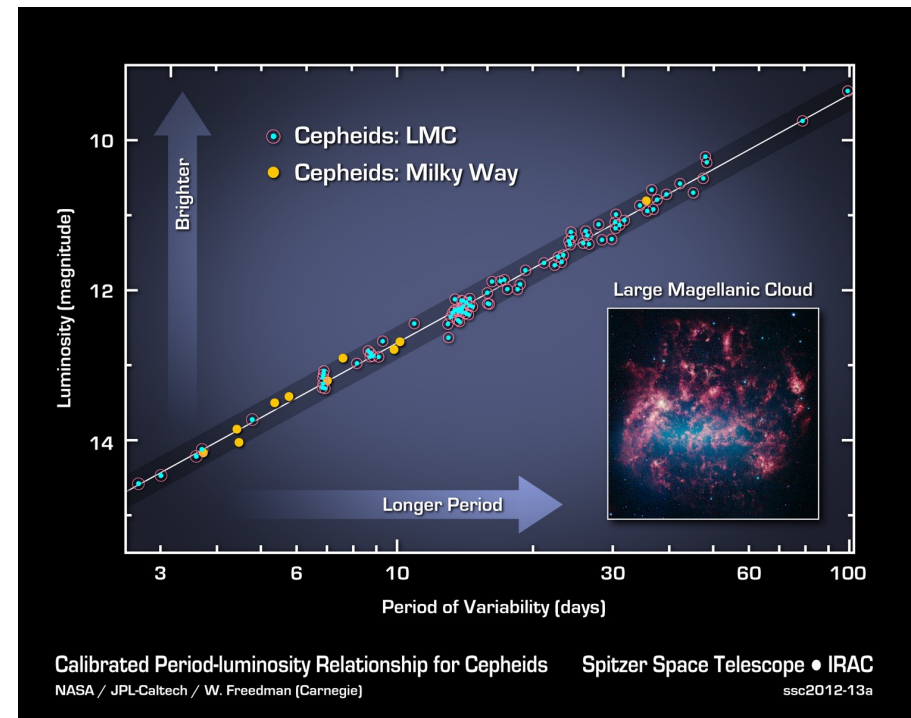
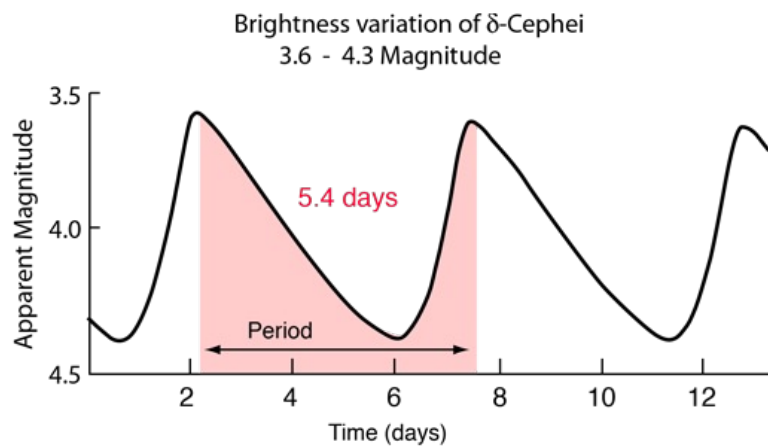
Cosmic Distance Ladder



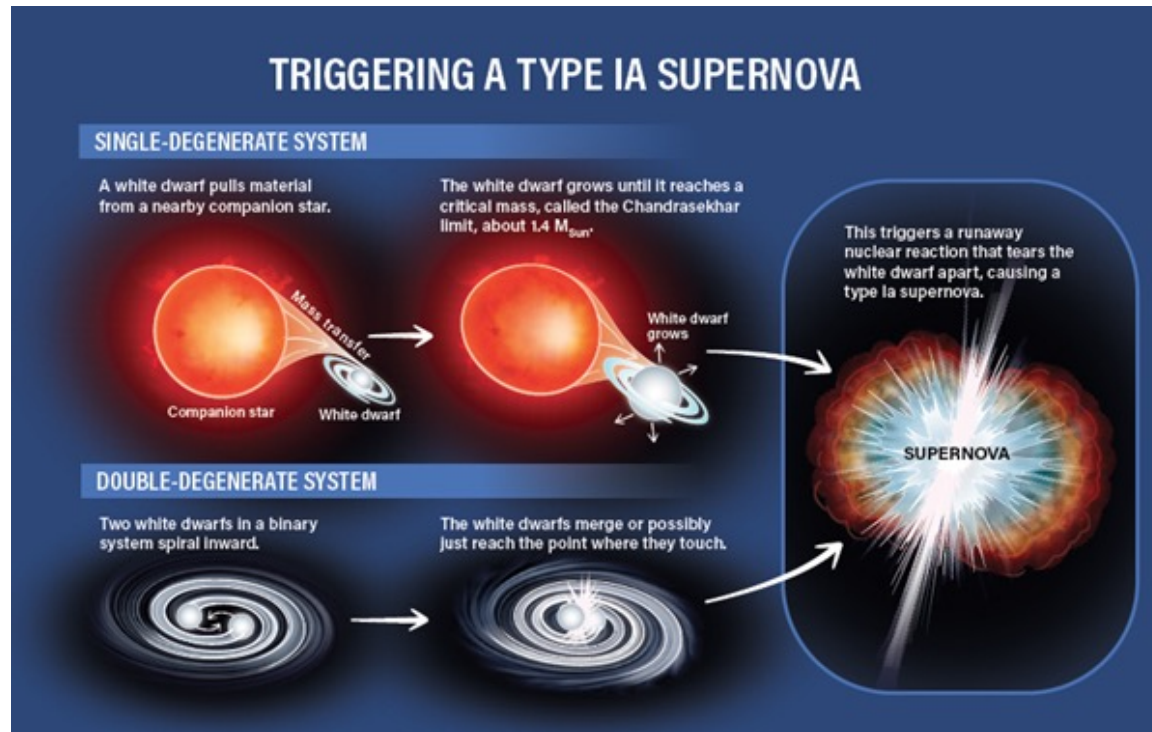
Geometry



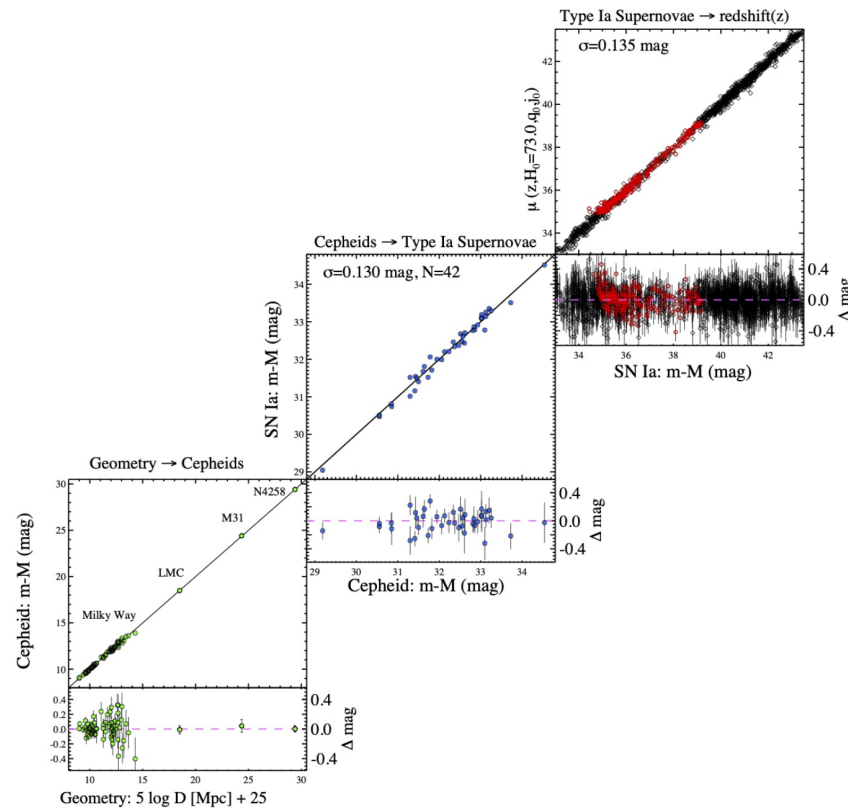
Cepheids



Type Ia Supernovae



Cosmic Distance Ladder: Expansion Rate!



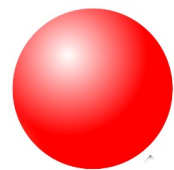
Thus, the Universe does in fact expand!

Two Predictions of an Expanding Universe

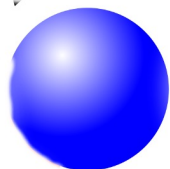
- 1) A Singularity (often called 'The Big Bang') [Hence, we can calculate age/size.]
 - a) Inflation? Ekpyrotic Universe? Quantum Gravity? Chaotic BKL Conditions? Other Jargon?
Observationally Unknown
- 2) Hot Big Bang (When the Universe was 1 second old)
 - a) Big Bang Nucleosynthesis (BBN) : Light Element Production
 - b) Cosmic Microwave Background (CMB) : Thermal Afterglow

Let's Explore the Predictions of the Hot Big Bang!

Big Bang Nucleosynthesis (BBN)



Protons (p)



Neutrons (n)



Electrons (e)



Neutrinos (ν)



Photons (γ)

At 1 second Weak Interactions

$$\nu_e + \bar{\nu}_e \leftrightarrow e^+ + e^-$$

$$e^- + \bar{\nu}_e \leftrightarrow e^- + \bar{\nu}_e$$

$$n + \nu_e \leftrightarrow p^+ + e^-$$

$$n + e^+ \leftrightarrow p^+ + \bar{\nu}_e$$

Turn off

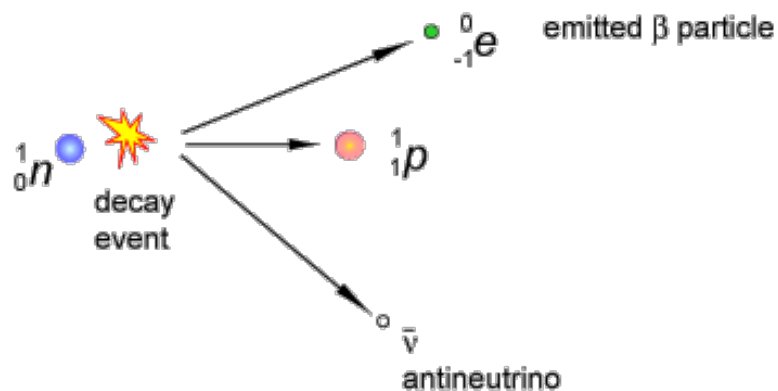
Big Bang Nucleosynthesis (BBN)

$$t \gtrsim 100 \text{ sec}$$

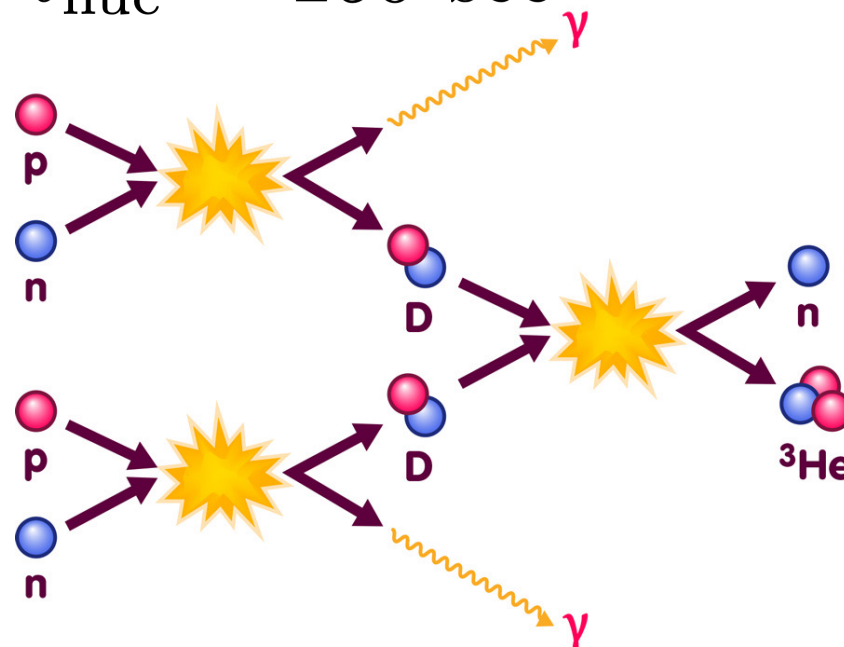
$$X_n(t_{\text{nuc}}) \sim \frac{1}{8}$$

$$t_{\text{nuc}} \simeq 250 \text{ sec}$$

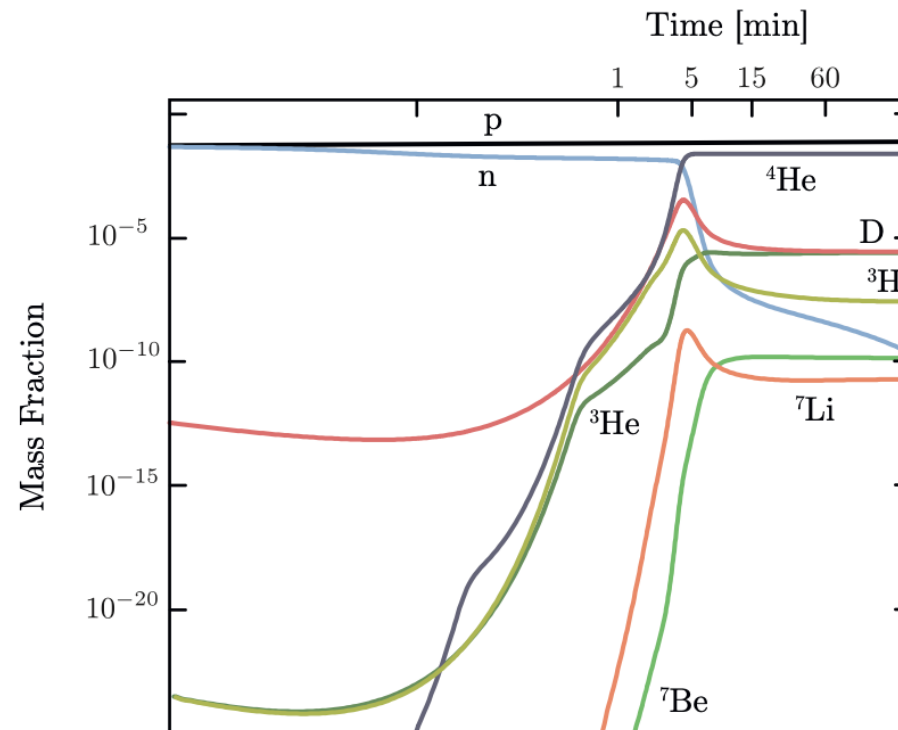
Beta Decay of a Neutron



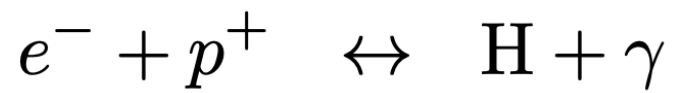
$$\frac{4n_{\text{He}}}{n_{\text{H}}} \sim \frac{1}{4}$$



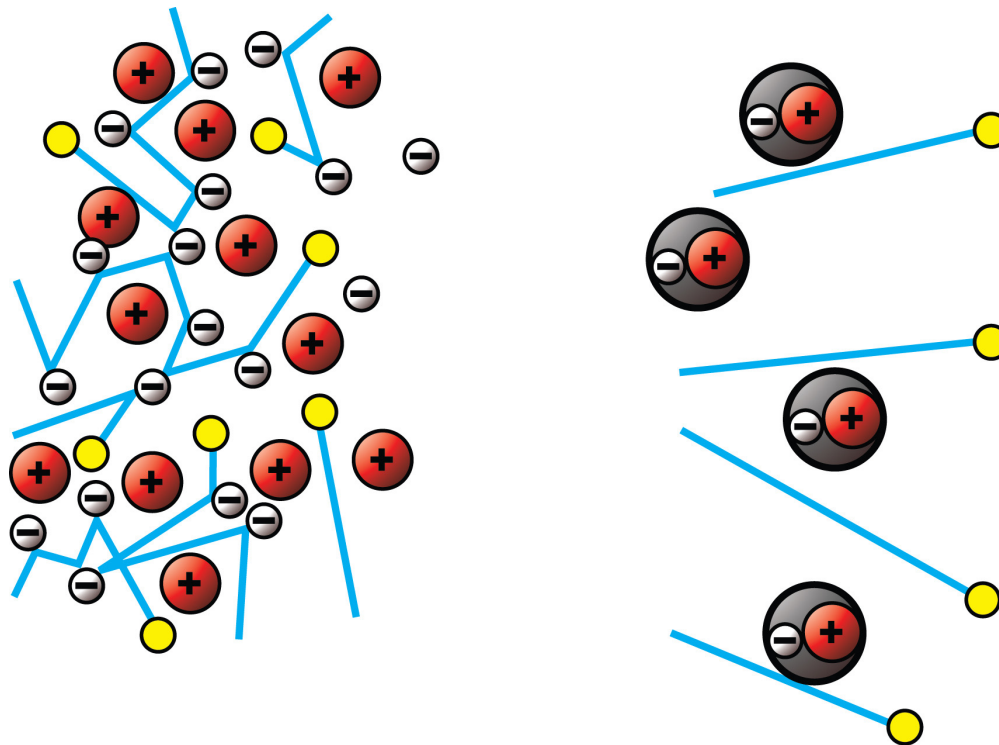
Big Bang Nucleosynthesis (BBN)



Recombination

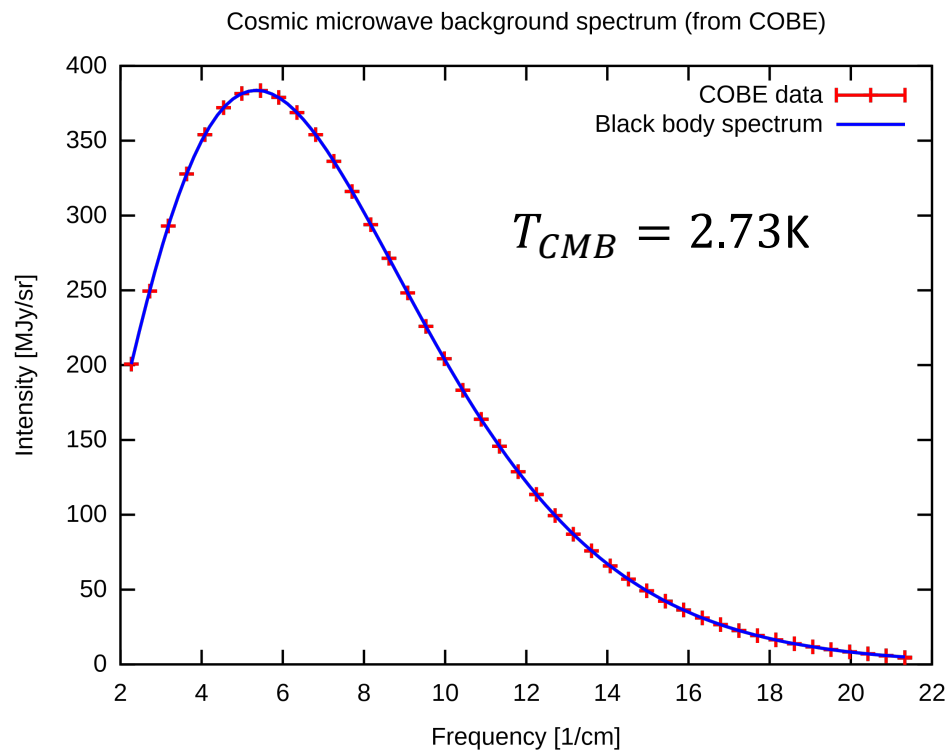


380 kyr

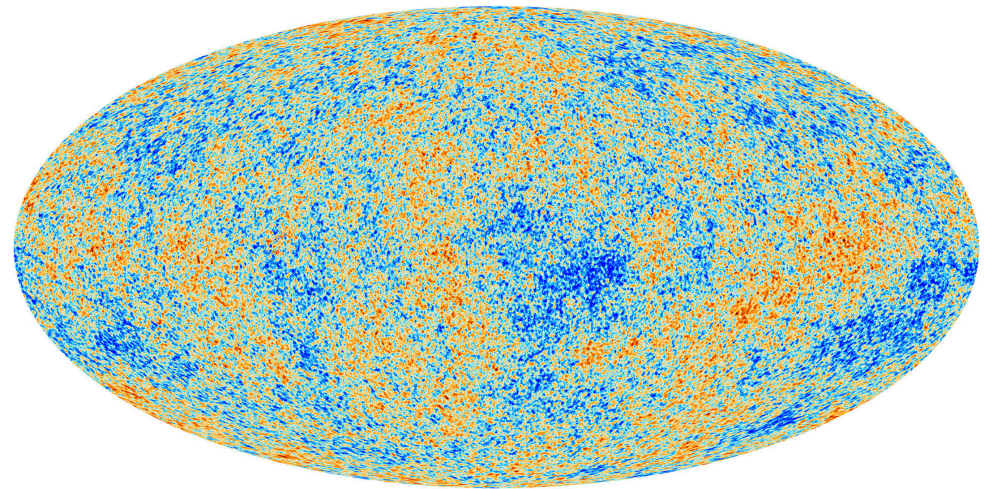


Free Streaming Photons!

Cosmic Microwave Background (CMB)



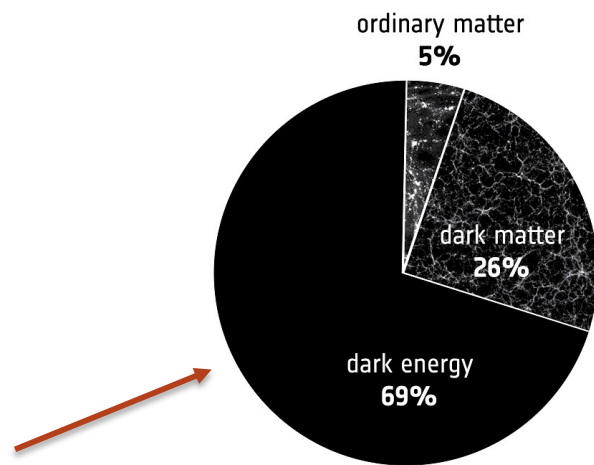
Very Isotropic (Uniform in All Directions!)



First discovered in 1965 by Penzias and Wilson
COBE Measured Spectrum 1993

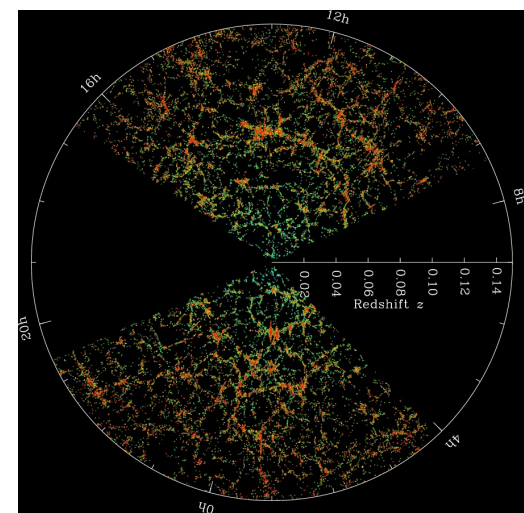
CMB Temperature Fluctuations

Energy Budget of Universe:



Current Accelerated Expansion!

Agree with Galaxy Density Fluctuations!



How did these temperature and density fluctuations arise?
Fundamentally, we still do not know.

What are the Fluctuations?

What are the slight fluctuations seen in maps of the cosmic background radiation?

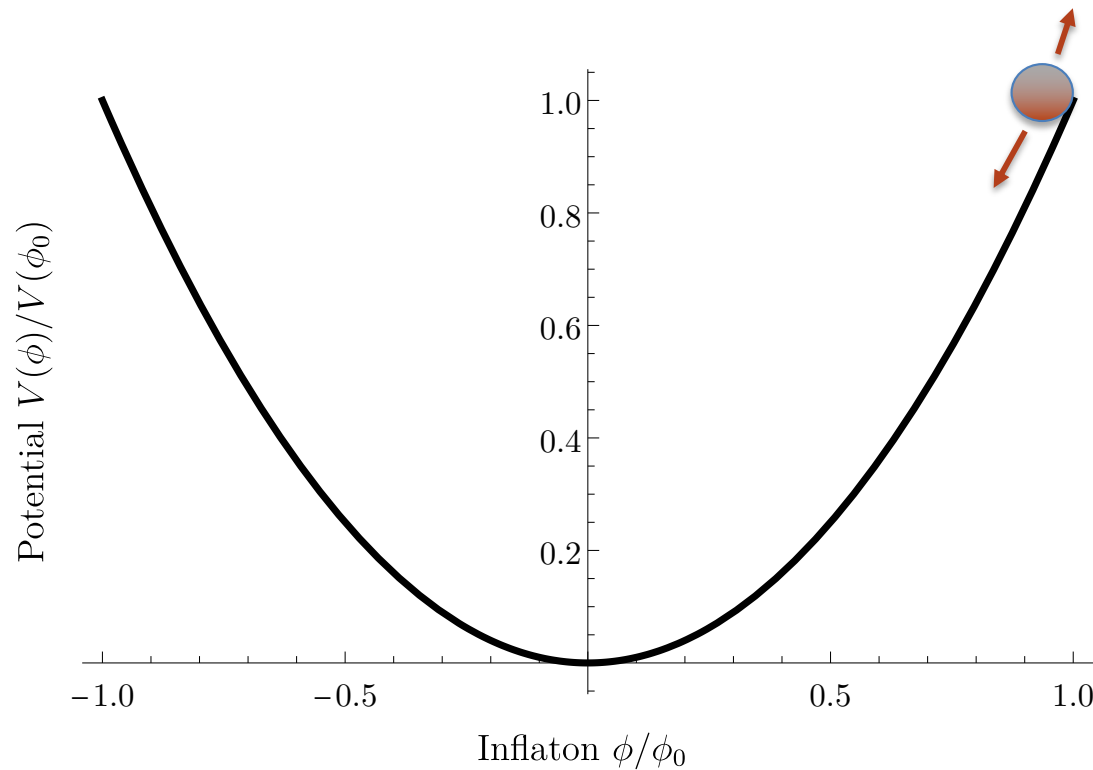
- A) Errors in the map
- B) Variations in the instrument's sensitivity
- C) The beginning of the formation of galaxies and clusters of galaxies
- D) Dark matter
- E) None of the above

Three Problems with the Expanding Universe!

- 1) The Horizon Problem
- 2) The Flatness Problem
- 3) Magnetic Monopole Problem

One Hypothesis for the beginning: Inflation

A period of accelerated expansion in the early Universe



Solves:

- 1) The Horizon Problem
- 2) The Flatness Problem
- 3) Magnetic Monopole Problem

Quantum Inflaton Fluctuations Seed
Density Perturbations